A Five-Factor Asset Pricing Model

Fama and French (2015), *Journal of Financial Economics* 116, 1–22

Two ways to choose a factor

Twenty-two years separate FF (1993) from FF (2015). The gap is not idleness. It marks a change in how the factors are justified.

1993: chosen from the data

Size and B/M were, in FF's words, "the two well-known patterns in average returns left unexplained by the CAPM." Start from observed failures and build factors to span them.

2015: chosen from an identity

Many anomaly variables could have been added. FF anticipate the objection, "it is reasonable to ask why we choose profitability and investment," and answer from the dividend discount model, before any regression runs.

Derived, not discovered. That is the standard; the talk applies it to all five factors, including the old ones.

The starting point is a tautology

$$m_t = \sum_{\tau=1}^{\infty} \frac{E(d_{t+\tau})}{(1+r)^{\tau}} \tag{1}$$

Which quantity is the unknown?

Usually r is an *input*: take a required return from some model, compute the price. Here it is the *output*: take the observed price and the dividend forecasts, and solve for the rate that reconciles them. That is an internal rate of return.

Solved that way, (1) can never be contradicted by data. It does not restrict r; it defines r. Feed it any price at all, including a wrong one, and out comes the rate that rationalises it.

Which is exactly what FF need. Every prediction drawn from (1) holds "whether the price is rational or irrational." The cost is that an identity proves nothing on its own.

Clean surplus turns dividends into earnings and investment

Book equity absorbs everything not paid out:

$$B_{t+\tau} = B_{t+\tau-1} + Y_{t+\tau} - D_{t+\tau} \quad \Longrightarrow \quad D_{t+\tau} = Y_{t+\tau} - dB_{t+\tau}$$

Substituting into (1) at firm level (Miller–Modigliani, 1961) and dividing by book equity:

$$\frac{M_t}{B_t} = \frac{\sum_{\tau=1}^{\infty} E(Y_{t+\tau} - dB_{t+\tau})/(1+r)^{\tau}}{B_t} \tag{3}$$

Y is total equity earnings; $dB_{t+\tau} = B_{t+\tau} - B_{t+\tau-1}$ is investment.

A bookkeeping substitution, not an economic assumption, so (3) inherits the tautological status of (1).

Four comparative statics, of which FF report three

Five quantities, M_t, B_t, $E(Y)$, $E(dB)$, r, of which r is solved for. Four free inputs; fix three, vary one: $\binom{4}{1} = 4$.

Vary	Holding fixed	Effect on r
M_t up (i.e. B/M down)	B_t, $E(Y)$, $E(dB)$	down
$E(Y)$ up	M_t, B_t, $E(dB)$	up
$E(dB)$ up	M_t, B_t, $E(Y)$	down
B_t up	*M_t, $E(Y)$, $E(dB)$*	*redundant with row 1*

Third row: higher expected investment means smaller expected payouts $E(Y - dB)$ at the same price, so r must *fall*. This is why CMA is long *low* investment.

The ceteris paribus is not innocent: retained earnings normally raise future earnings, and row 3 shuts that channel off.

The reframing that drives everything else

FF's own summary of (3)

B_t/M_t is a noisy proxy for expected return, because M_t also responds to forecasts of earnings and investment.

The three-factor model has a factor for one determinant of r and none for the other two. That is the whole case for RMW and CMA.

Two consequences carried through the paper:

- Predictions are conditional. Eq. (3) does not say B/M, OP and Inv effects appear without controls, and the three characteristics are correlated (FF 1995).
- If B/M works partly by proxying for the other two, adding them should shrink what it contributes. Foreshadows the redundancy result.

The model, and the factor the identity never implied

$$R_{it} - R_{Ft} = a_i + b_i(R_{Mt} - R_{Ft}) + s_i\mathrm{SMB}_t + h_i\mathrm{HML}_t + r_i\mathrm{RMW}_t + c_i\mathrm{CMA}_t + e_{it} \qquad (5)$$

Null hypothesis: $a_i = 0$ for all i.

Eq. (3) implies B/M, profitability and investment. It does not imply size. FF's escape clause: variables "not explicitly linked to this decomposition, such as Size and momentum" must work by improving forecasts of the others, or by capturing horizon effects.

So SMB survives on precedent and empirics, the 1993 standard rather than the 2015 one. It nevertheless earns its place: its premium is not spanned by the other four, and every failure in the paper is a small-stock failure.

How a characteristic becomes a factor

Each June, sort independently: Size at the NYSE median, the second variable at the 30th and 70th NYSE percentiles. Six value-weighted cells per sort, held for twelve months. Three sorts, one per characteristic.

Size × B/M

	Low	Neutral	High
Small	SL	SN	SH
Big	BL	BN	BH

$$\mathrm{HML} = \frac{SH+BH}{2} - \frac{SL+BL}{2}$$

Size × OP

	Weak	Neutral	Robust
Small	SW	SN	SR
Big	BW	BN	BR

$$\mathrm{RMW} = \frac{SR+BR}{2} - \frac{SW+BW}{2}$$

Size × Inv

	Low	Neutral	High
Small	SC	SN	SA
Big	BC	BN	BA

$$\mathrm{CMA} = \frac{SC+BC}{2} - \frac{SA+BA}{2}$$

Teal is the long leg, coral the short leg, gray is used only by size factors. Note the third grid flips: CMA is long *low* investment.

Each sort also yields a size factor, small row minus big row. SMB is the average of the three.

Why double sort? Because $S(B) \propto H(L)$

Size and B/M are correlated. So a univariate high-minus-low carries a size tilt, a univariate small-minus-big carries a value tilt, and the two factors come out entangled.

The fix is visible in the second way of writing HML:

$$\text{HML} = \frac{(SH - SL) + (BH - BL)}{2}$$

a value factor computed *within* small stocks, averaged with one computed *within* big stocks. The size exposure nets out.

Value weighting makes the univariate version worse still. It would be dominated by megacaps, approximating HML_B alone (0.21, $t = 1.69$) and discarding HML_S (0.53, $t = 4.05$).

Caveat: this nets out size *between* bins, not within them, and FF concede HML is not neutral with respect to OP or Inv at all.

The factors themselves (Table 4, 2×3, % per month)

	Mean	Std dev	t
RMRF	0.50	4.49	2.74
SMB	0.29	3.07	2.31
HML	0.37	2.88	3.20
RMW	0.25	2.14	2.92
CMA	0.33	2.01	4.07

- **The premiums live in small stocks.** $\mathrm{HML}_S = 0.53$ versus $\mathrm{HML}_B = 0.21$ ($t = 1.69$). CMA_S is 4.64–5.49 standard errors from zero; CMA_B only 1.03–2.00.
- HML and CMA correlate about 0.70, since value firms invest little. The first quantitative hint of the redundancy result, three sections early.

The left-hand side: FF's "playing field"

Testing $a_i = 0$ needs portfolios to test it *on*. FF want wide spreads in the four characteristics, so they sort stocks on those characteristics directly.

Six sets, value-weighted, NYSE breakpoints, July 1963 to December 2013:

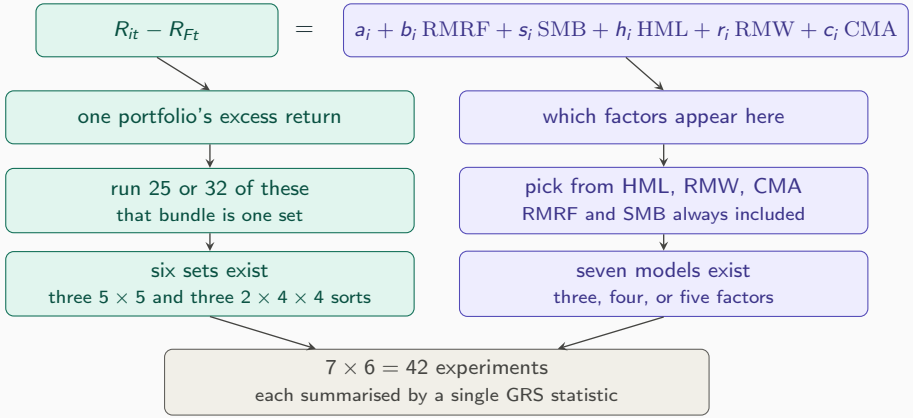
- Three 5×5 sorts: Size crossed with B/M, OP, or Inv (25 each)
- Three $2 \times 4 \times 4$ sorts: Size plus two of the others (32 each)

Each set is a separate experiment: run 25 or 32 regressions, then test those intercepts jointly.

Why the second kind: the characteristics are correlated, so bivariate sorts do not isolate the effects (3) predicts conditionally. Adding a control costs diversification.

The villain appears before any regression. Microcap $\times$ highest Inv earns 0.35% per month, the lowest cell in the matrix. FF flag in advance that it will be "lethal."

The test design



GRS (Gibbons, Ross and Shanken, 1989) tests $H_0 : \alpha_1 = \cdots = \alpha_N = 0$ jointly. FF also rebuild the factors three ways, so the full grid is $7 \times 6 \times 3$.

The scoreboard

	$A\|a_i\|/A\|r_i\|$	Variance left unexplained
CAPM	1.26–1.55	n/a
Three-factor	0.54–0.68	mostly > 0.50
Five-factor	0.42–0.54	0.06–0.29

GRS rejects every model on every set. FF pivot: models are "simplified propositions about expected returns that are rejected in tests with power." Use GRS to rank, not to pass.

Biggest gain is on the 32 Size–OP–Inv portfolios (6.9–8.2 bp per month), exactly where the identity says it should be.

HML is redundant

The five-factor model beats everything except the four-factor model that drops HML, which matches it on every measure including GRS.

Table 6 regresses each factor on the other four. Four of five have intercepts more than three standard errors from zero. HML's is -0.04 ($t = -0.47$): its premium is absorbed by the CMA slope (1.04, $t = 23.03$) and the RMW slope (0.23, $t = 5.36$).

The distinction

HML still earns 0.37% per month with $t = 3.20$. What it lacks is alpha against the other four. Drop it for pricing; keep it for attribution. Hence HML^{O}, which preserves h_i but has a mean near zero.

FF caution this "may be specific to this sample."

Where the model breaks

Small stocks with negative RMW and CMA loadings: returns behaving like firms that invest a lot despite low profitability.

Portfolio	Three-factor a	Five-factor a
Microcap, lowest B/M (T.7)	-0.49 (-5.18)	-0.29 (-3.31)
Microcap, highest Inv (T.10)	-0.48 (-7.19)	-0.35 (-5.30)
Small, low OP $\times$ high Inv (T.11)	-0.87 (-8.45)	-0.47 (-5.89)

With $E[\mathrm{RMW}] > 0$ and $E[\mathrm{CMA}] > 0$, negative loadings already push the predicted return *down*. Realised returns are lower still: direction right, magnitude wrong. The loadings absorb roughly 40% of the three-factor anomaly and stop.

Any one of these rejects the model for its whole set on Bonferroni's inequality.

The result FF leave unexplained

Big stocks with the same loading signature get positive intercepts.

	Small	Big
Low OP $\times$ high Inv (T.11)	-0.47 (-5.89)	$+0.12$ (1.37)
Low B/M $\times$ high Inv (T.A3)	-0.20 (-4.18)	$+0.37$ (5.39)
Low B/M $\times$ low OP (T.A2)	-0.33 (-3.49)	$+0.34$ (2.01)

FF's challenge to behavioural stories

"The unexplained average returns of big stocks that invest a lot despite low profitability are positive." Same characteristics, opposite-signed mispricing. A factor model cannot produce this, and neither can a story about investors overpaying for glamorous cash-burning firms.

Back to the standard we set

The new factors pass

Eq. (3) nominated profitability and investment before any regression ran. Both carry premiums, both improve every metric, and the improvement is largest exactly where the identity says it should be.

The old ones fail, in opposite directions

- B/M *is* implied by (3), and turns out redundant in the data: the identity's own headline variable drops out.
- Size is *not* implied by (3), and turns out indispensable: unspanned by the other four, and the site of every failure in the paper.

And GRS rejects everywhere. The identity tells you which variables belong. It never tells you they are priced, and says nothing about where the model breaks.

Open question: does the test design have teeth?

Unless it is out of sample, is it circular?

Against circularity. The intercept is a free parameter. OLS forces the *residual* mean to zero, not the *return* mean; loadings fit covariation, which does not pin down means. A circular procedure could not produce -0.47 with $t = -5.89$. The model demonstrably fails.

For the concern. All in-sample, one dataset, 606 months. Test portfolios are "finer versions of the sorts that produce the factors," in FF's own words. They promise "more hostile territory" in FF (2014).

Where it bites hardest. The redundancy result is precisely a within-sample spanning claim, estimated on a window ending December 2013, immediately before HML's worst stretch on record.

Better, and still wrong.

Backup: two readings of the zero-intercept hypothesis

Modest (Huberman–Kandel, 1987)

The mean-variance-efficient tangency portfolio is spanned by the riskfree asset, the market, SMB, HML, RMW and CMA. A statement about the efficient frontier, with no risk story attached.

Ambitious (Merton, 1973)

Up to four unspecified state variables generate premiums the market factor misses. But Size, B/M, OP and Inv are not state variables, and the factors are not mimicking portfolios, just diversified portfolios giving different exposure combinations (Fama, 1996).

With state variables unspecified and no mimicking claim, the second reduces to spanning plus a gesture.

Backup: what is a value stock?

Purely mechanical: one end of the B/M sort. No profitability or quality content; those enter separately.

	Low B/M	High B/M
Price relative to book	expensive	cheap
Label	growth	value
Table 8 investment (microcaps)	0.29	0.04

Modern caveat. Accounting expenses R&D and brand investment rather than capitalising it, so book equity understates the asset base for intangible-intensive firms, which are disproportionately the largest ones. This pushes mega-caps toward low B/M mechanically and degrades B/M as a sorting variable.

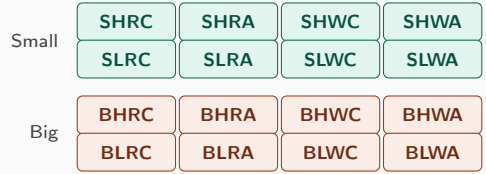
Backup: the other two construction schemes

2×2 sorts

Second variable split at the NYSE median too, so all stocks are used. Lower σ, smaller means, t-statistics essentially unchanged.

$2 \times 2 \times 2 \times 2$ sorts

One sort on all four variables at NYSE medians. Sixteen cells, coded Size, B/M, OP, Inv. Reallocates premiums: RMW 0.17 $\rightarrow$ 0.25, CMA 0.22 $\rightarrow$ 0.14.



Each factor is an 8-vs-8 split along one letter position:

SMB: 1st, S – B (teal – coral)
HML: 2nd, H – L
RMW: 3rd, R – W
CMA: 4th, C – A

A defect: HML and RMW correlate 0.63 here, because SLWC and SLWA, the two highest-variance cells, sit short in both. FF call this artificial.

Backup: slopes do not track characteristics

- Microcap extreme-growth stocks carry the most negative RMW slope in the matrix (-0.58, $t = -13.26$), yet Table 8 says they are *more* profitable than other microcap portfolios.
- Low OP per se is not the problem: the microcap lowest-OP portfolio has a huge negative RMW slope (-0.67, $t = -17.70$) but only modest CMA exposure and $a = -0.10$ ($t = -1.28$).

Multivariate slopes measure marginal effects; univariate characteristics need not agree. Hence FF's careful "behave like" phrasing throughout.

In the $2 \times 4 \times 4$ Size–OP–Inv sorts the two do line up, which is why FF say there is "less ambiguity" there.

Backup: two gaps the identity does not close

1. **Horizon.** r is one rate over an infinite stream; the tests use one-month returns. FF assume the short-horizon expected return is "positively related" to the IRR. Monotonicity is what they need, and a flat term structure is offered as one sufficient case. They concede Size and momentum may work through "horizon effects in the term structure of expected returns."
2. **Expectations.** Eq. (3) is about expected future profitability and investment. The sorts use lagged realisations.

Also: investment in (3) is growth in *book equity*; the sorts use growth in *total assets*. FF call the substitution innocuous.

Backup: loose ends FF hand to the literature

- Is high investment *alone* the problem for small stocks, rather than the investment/profitability combination?
- Investment in the top quintile runs from 43% of assets for megacaps to 71% for microcaps. How much is new issues and stock-financed mergers (Loughran–Ritter; Ikenberry et al.)?
- Do the slopes line up better with *future* characteristics than past ones, since pricing is forward looking?
- Does the HML redundancy survive pre-1963 or outside the US?

Closest related work: Hou, Xue and Zhang (2012). FF object that they use value-weighted univariate sorts, dominated by big stocks, and "the most serious problems of asset pricing models are in small stocks."